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Ingredient container

Abstract

Ingredient containers for use in a beverage carbonation system are provided. The ingredient container includes a container body having a hollow interior and an opening leading into the hollow interior. The ingredient container also includes a cap positioned over the opening in the container body and including an inlet port and an outlet port. A cross-section of the cap extends substantially perpendicular to a central axis of each of the inlet port and the outlet port, and the cross-section of the cap has a shape that is a substantially circular triangle.

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9743801	12/2016	Leuzinger et al.	N/A	N/A
9745120	12/2016	Abegglen et al.	N/A	N/A
9745185	12/2016	Klopfenstein et al.	N/A	N/A
9751054	12/2016	Jin et al.	N/A	N/A
9754437	12/2016	Deo et al.	N/A	N/A
9770129	12/2016	Remo et al.	N/A	N/A
9783403	12/2016	Tansey, Jr.	N/A	N/A
9783405	12/2016	Olson et al.	N/A	N/A
9788681	12/2016	Perentes et al.	N/A	N/A
9790076	12/2016	Novak et al.	N/A	N/A
9796506	12/2016	Meager	N/A	N/A
9801500	12/2016	Ven Der Woning	N/A	N/A
9809437	12/2016	Tansey, Jr.	N/A	N/A
9811704	12/2016	Kaaser	N/A	N/A
9821951	12/2016	Estabrook et al.	N/A	N/A
9821992	12/2016	Rudick et al.	N/A	N/A
9854935	12/2017	Danieli et al.	N/A	N/A
9889239	12/2017	Michaels et al.	N/A	N/A
9889966	12/2017	Medeiros et al.	N/A	N/A
9896322	12/2017	Hecht	N/A	N/A
9897220	12/2017	Cohen et al.	N/A	N/A
9907432	12/2017	Tanner et al.	N/A	N/A
9918586	12/2017	Smith et al.	N/A	N/A
9957145	12/2017	Cohen et al.	N/A	N/A
9974410	12/2017	Ferrier	N/A	N/A
9980596	12/2017	Rognon et al.	N/A	N/A
9981801	12/2017	Ozanne et al.	N/A	N/A
9999315	12/2017	Crarer et al.	N/A	N/A
9999316	12/2017	Ye et al.	N/A	N/A
10000370	12/2017	Bethuy et al.	N/A	N/A
10007397	12/2017	Besson et al.	N/A	N/A
10017372	12/2017	Bethuy et al.	N/A	N/A
10022011	12/2017	Norton et al.	N/A	N/A
10028614	12/2017	Perentes et al.	N/A	N/A
10034573	12/2017	Flick et al.	N/A	N/A
10046903	12/2017	Evans et al.	N/A	N/A
10046904	12/2017	Evans et al.	N/A	N/A
10051988	12/2017	Gordon et al.	N/A	N/A
10058826	12/2017	Cohen et al.	N/A	N/A
10064513	12/2017	Rehfuss	N/A	N/A
10070751	12/2017	Magniet et al.	N/A	N/A
10076208	12/2017	Castellani et al.	N/A	N/A
10080461	12/2017	Bugnano et al.	N/A	N/A
10093530	12/2017	Mackey et al.	N/A	N/A

10099443	12/2017	Evans et al.	N/A	N/A
10106392	12/2017	Peirsman et al.	N/A	N/A
10117539	12/2017	Rognon et al.	N/A	N/A
10117540	12/2017	De Vreede et al.	N/A	N/A
10130211	12/2017	Bugnano et al.	N/A	N/A
10131528	12/2017	Webster et al.	N/A	N/A
10131529	12/2017	Jersey et al.	N/A	N/A
10136755	12/2017	Talon	N/A	N/A
10143782	12/2017	Yurek et al.	N/A	N/A
10143978	12/2017	Tipton	N/A	N/A
10149569	12/2017	Preshel	N/A	N/A
10155647	12/2017	Foster et al.	N/A	N/A
10159376	12/2017	Dovat et al.	N/A	N/A
10160575	12/2017	Ray	N/A	N/A
10165892	12/2018	Lafosse	N/A	N/A
10189614	12/2018	Pruiett	N/A	N/A
10193411	12/2018	Tajima et al.	N/A	N/A
10201171	12/2018	Gordon et al.	N/A	N/A
10201785	12/2018	Cohen et al.	N/A	N/A
10206533	12/2018	Pirone	N/A	N/A
10211438	12/2018	Ohashi et al.	N/A	N/A
10213033	12/2018	Bratsch et al.	N/A	N/A
10213752	12/2018	Shalev	N/A	N/A
10214018	12/2018	Nozawa et al.	N/A	N/A
10227226	12/2018	Jersey et al.	N/A	N/A
10229401	12/2018	Yoakim	N/A	N/A
10231569	12/2018	Perentes et al.	N/A	N/A
10233002	12/2018	Baenninger et al.	N/A	N/A
10239669	12/2018	Ayriss et al.	N/A	N/A
10258186	12/2018	Rivera	N/A	N/A
10280060	12/2018	Van Opstal et al.	N/A	N/A
10294020	12/2018	Nordqvist et al.	N/A	N/A
10307718	12/2018	Waisman	N/A	N/A
10329134	12/2018	Olson et al.	N/A	N/A
10334871	12/2018	Van De Sluis et al.	N/A	N/A
10336597	12/2018	Griscik et al.	N/A	N/A
10343885	12/2018	Novak et al.	N/A	N/A
10349773	12/2018	Segiet et al.	N/A	N/A
10350561	12/2018	Dushine et al.	N/A	N/A
10358269	12/2018	Cervený	N/A	N/A
10364089	12/2018	Daniels et al.	N/A	N/A
10365141	12/2018	Freiburger et al.	N/A	N/A
10370235	12/2018	Pellaud	N/A	N/A
10377540	12/2018	Borgardt et al.	N/A	N/A
10377620	12/2018	Makino et al.	N/A	N/A
10384839	12/2018	Yamaguchi	N/A	N/A
10398254	12/2018	Tinkler et al.	N/A	N/A
10399769	12/2018	Talon et al.	N/A	N/A
10399838	12/2018	Green	N/A	N/A
10399839	12/2018	Knoll et al.	N/A	N/A

10405690	12/2018	Tentorio	N/A	N/A
10405691	12/2018	Hesselbrock et al.	N/A	N/A
10414557	12/2018	Skillin et al.	N/A	N/A
10414642	12/2018	Melville, Jr. et al.	N/A	N/A
10433668	12/2018	Merali et al.	N/A	N/A
10433671	12/2018	Surface	N/A	N/A
10442591	12/2018	Rognard et al.	N/A	N/A
10455968	12/2018	Singer	N/A	N/A
10455973	12/2018	Dollner et al.	N/A	N/A
10455974	12/2018	Talon	N/A	N/A
10456539	12/2018	Hearn et al.	N/A	N/A
10456757	12/2018	Blichmann	N/A	N/A
10457450	12/2018	Rios	N/A	N/A
10470605	12/2018	Ergican et al.	N/A	N/A
10479669	12/2018	Kim et al.	N/A	N/A
10485374	12/2018	Lo Faro et al.	N/A	N/A
10486953	12/2018	Pellaud et al.	N/A	N/A
10488097	12/2018	Nachawati et al.	N/A	N/A
10494246	12/2018	Hecht et al.	N/A	N/A
10506896	12/2018	Ven Der Woning	N/A	N/A
10507958	12/2018	Hashimoto et al.	N/A	N/A
10513424	12/2018	Tansey, Jr.	N/A	N/A
10518938	12/2018	Suzuki et al.	N/A	N/A
10518942	12/2018	Seibert et al.	N/A	N/A
10519020	12/2018	Ozawa et al.	N/A	N/A
10524617	12/2019	Perrin et al.	N/A	N/A
10526186	12/2019	Kuboi et al.	N/A	N/A
10526192	12/2019	Holley et al.	N/A	N/A
10543977	12/2019	Brockman et al.	N/A	N/A
10548430	12/2019	Guard et al.	N/A	N/A
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10562700	12/2019	Weijers et al.	N/A	N/A
10568452	12/2019	Fin et al.	N/A	N/A
10595549	12/2019	Van De Sluis et al.	N/A	N/A
10595668	12/2019	Tinkler et al.	N/A	N/A
10596305	12/2019	Roberts et al.	N/A	N/A
10604310	12/2019	Kutsuzawa et al.	N/A	N/A
10604398	12/2019	Smeller et al.	N/A	N/A
10618705	12/2019	Laible	N/A	N/A
10631686	12/2019	Abdo et al.	N/A	N/A
10647564	12/2019	Showalter	N/A	N/A
10654700	12/2019	Hecht	N/A	N/A
10674857	12/2019	Lyons et al.	N/A	N/A
10674863	12/2019	Sevcik et al.	N/A	N/A
10676336	12/2019	Makino et al.	N/A	N/A
10682007	12/2019	Fischer	N/A	N/A
10682593	12/2019	Baird	N/A	N/A
10702835	12/2019	Tran et al.	N/A	N/A
10702838	12/2019	Chaussin et al.	N/A	N/A
10703618	12/2019	Ziesel	N/A	N/A

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10710864	12/2019	Jangbarwala et al.	N/A	N/A
10717567	12/2019	Sakamoto et al.	N/A	N/A
10717637	12/2019	Pellaud et al.	N/A	N/A
10743707	12/2019	Bugnano et al.	N/A	N/A
10759594	12/2019	Mills et al.	N/A	N/A
10765254	12/2019	Iotti et al.	N/A	N/A
10766756	12/2019	Gatipon et al.	N/A	N/A
10772460	12/2019	Accursi	N/A	N/A
10780408	12/2019	Schöb et al.	N/A	N/A
10791752	12/2019	Siegel et al.	N/A	N/A
10793346	12/2019	Bartoli et al.	N/A	N/A
10800581	12/2019	Berroa Garcia	N/A	N/A
10807049	12/2019	Abdo et al.	N/A	N/A
10807853	12/2019	Balstad et al.	N/A	N/A
10813501	12/2019	Helf et al.	N/A	N/A
10820741	12/2019	Byun et al.	N/A	N/A
10820744	12/2019	Rubin et al.	N/A	N/A
10820745	12/2019	Zwicker et al.	N/A	N/A
10820746	12/2019	Noth	N/A	N/A
10827875	12/2019	Noth	N/A	N/A
10828586	12/2019	Simpson et al.	N/A	N/A
10829359	12/2019	Von Kraus et al.	N/A	N/A
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10843142	12/2019	Waggoner et al.	N/A	N/A
10843849	12/2019	Berge	N/A	N/A
10843866	12/2019	Cafaro et al.	N/A	N/A
10846975	12/2019	Tansey et al.	N/A	N/A
10849451	12/2019	Su	N/A	N/A
10849454	12/2019	Gordon et al.	N/A	N/A
10869572	12/2019	Blatt	N/A	N/A
10870566	12/2019	Green et al.	N/A	N/A
10882728	12/2020	Hong et al.	N/A	N/A
10883072	12/2020	Hong et al.	N/A	N/A
10893773	12/2020	Standaar et al.	N/A	N/A
10894639	12/2020	Pruiett	N/A	N/A
10894706	12/2020	Iotti et al.	N/A	N/A
10898026	12/2020	Fin	N/A	N/A
10899600	12/2020	Frieburger et al.	N/A	N/A
10905287	12/2020	Tu et al.	N/A	N/A
10906013	12/2020	Cohen et al.	N/A	N/A
10918239	12/2020	Hartmann et al.	N/A	N/A
10919752	12/2020	Breault	N/A	N/A
10925433	12/2020	Hansen et al.	N/A	N/A
10926945	12/2020	Kennedy et al.	N/A	N/A
10926995	12/2020	Seay et al.	N/A	N/A
10940494	12/2020	Romanov et al.	N/A	N/A
10945554	12/2020	Lo Faro et al.	N/A	N/A
10945557	12/2020	Nishimura et al.	N/A	N/A
10947485	12/2020	Min et al.	N/A	N/A

10952562	12/2020	Tanner et al.	N/A	N/A
10954043	12/2020	Taruno	N/A	N/A
10961027	12/2020	Laible	N/A	N/A
10966563	12/2020	Dubief et al.	N/A	N/A
10966564	12/2020	Rijskamp et al.	N/A	N/A
10973364	12/2020	Hesselbrock et al.	N/A	N/A
10981700	12/2020	Migas et al.	N/A	N/A
10989026	12/2020	Al-Gouhi et al.	N/A	N/A
10993575	12/2020	Krug et al.	N/A	N/A
10993576	12/2020	Fedorak et al.	N/A	N/A
10994980	12/2020	Jangbarwala et al.	N/A	N/A
11001490	12/2020	Headley et al.	N/A	N/A
11008206	12/2020	Pappas	N/A	N/A
11013363	12/2020	Alsudairi et al.	N/A	N/A
11021359	12/2020	Bissen et al.	N/A	N/A
11026539	12/2020	Zosimadis et al.	N/A	N/A
11033141	12/2020	Schlack	N/A	N/A
11039712	12/2020	Egli et al.	N/A	N/A
11040806	12/2020	Naumann et al.	N/A	N/A
11049354	12/2020	Yoakim	N/A	N/A
11053053	12/2020	Jordan	N/A	N/A
11059636	12/2020	Maeda	N/A	N/A
11064715	12/2020	Herbert et al.	N/A	N/A
11072521	12/2020	Walker	N/A	N/A
11078066	12/2020	Crackel et al.	N/A	N/A
11084007	12/2020	Adams	N/A	N/A
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11097236	12/2020	Alexander et al.	N/A	N/A
11109708	12/2020	Lecomte et al.	N/A	N/A
11110418	12/2020	Furman et al.	N/A	N/A
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11129491	12/2020	Park et al.	N/A	N/A
11147410	12/2020	Hachenberger et al.	N/A	N/A
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11166593	12/2020	Trakselis	N/A	N/A
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11192711	12/2020	Jarisch et al.	N/A	N/A
11194443	12/2020	Deo et al.	N/A	N/A
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11206941	12/2020	Abdo et al.	N/A	N/A
11208310	12/2020	Tansey et al.	N/A	N/A
11208313	12/2020	Conover et al.	N/A	N/A
11208314	12/2020	Peirsman et al.	N/A	N/A
11235267	12/2021	Santoiemmo	N/A	N/A
11242195	12/2021	Nordqvist et al.	N/A	N/A
11246326	12/2021	Feola	N/A	N/A

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11252976	12/2021	Popov et al.	N/A	N/A
11254491	12/2021	Krüger	N/A	N/A
11254586	12/2021	Santoiemmo	N/A	N/A
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11284736	12/2021	Ochoa et al.	N/A	N/A
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11292646	12/2021	Bai et al.	N/A	N/A
11292706	12/2021	Showalter	N/A	N/A
11292707	12/2021	Lecomte et al.	N/A	N/A
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11325760	12/2021	Alderson et al.	N/A	N/A
11325818	12/2021	Dahlberg et al.	N/A	N/A
11337542	12/2021	Kroos	N/A	N/A
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11345581	12/2021	Cook	N/A	N/A
11345583	12/2021	Aslam et al.	N/A	N/A
11370648	12/2021	Melville, Jr. et al.	N/A	N/A
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11479457	12/2021	Krüger et al.	N/A	N/A
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11745996	12/2022	Zbedlick	N/A	N/A
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2002/0130140	12/2001	Cote	N/A	N/A
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2011/0186535	12/2010	Meager	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/989,610, filed on Nov. 17, 2022, entitled “Ingredient Container,” which is hereby incorporated by reference herein in its entirety.

FIELD

(1) Ingredient containers used with beverage dispensing devices are provided.

BACKGROUND

(2) Conventional beverage dispensing devices operate to carbonate and/or flavor water. Some devices may mix carbonated water and a flavoring compound together in a machine and then dispense the resulting mixture into a receptacle. Unless the devices are thoroughly cleaned, this method can result in contamination occurring over time. Other devices rely on crushing, puncturing, and/or generally compromising flavoring containers in order to access the flavoring

compounds inside. These methods of breaching flavoring containers can result in splatter and mess, which, if not thoroughly cleaned, can result in similar contamination.

(3) Still other devices rely on carbonating water within a specialized container to be attached to the device, and from which the resulting beverage is served. The container can be pre-filled with water and/or flavoring, and then it can be secured to the devices and pressurized within the container and used to serve the resulting beverage. These devices, however, can create excess plastic waste, as specially adapted bottles must be produced to interface with the device.

(4) Accordingly, there remains a need for a better beverage dispensing device to improve on mess creation and waste production.

SUMMARY

(5) Ingredient containers for use with beverage dispensing systems are provided. Related apparatuses and techniques are also provided.

(6) In one embodiment, a container is provided and can include a container body defining a hollow interior, and a cap having an end wall with a first collar projecting therefrom and a second collar projecting therefrom. The first collar can have an inlet valve therein, and the second collar can have an outlet valve therein. The first and second collars can be spaced apart from one another. The end wall can further have first and second recesses surrounding the first and second collars. The first and second recesses can be formed in a surface of the end wall.

(7) The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the first and second collars and the first and second recesses together can define a figure-8 shaped feature. The container can also include first and second shoulder portions positioned on opposite sides of the end wall and projecting outward from the outward facing surface of the end wall. For example, each of the first and second recesses can have first and second curved sidewalls that extend partially around the first and second collars, respectively. In some aspects, each of the first and second recesses can have a third curved sidewall positioned opposite the first and second curved sidewalls. For example, the first and second recesses can be positioned on opposite sides of the first and second collars. For example, the cap can have a minor axis and a major axis, and wherein the cap is substantially symmetrical about the minor axis. In some aspects, the first and second collars can be aligned along the minor axis.

(8) In another embodiment, a container is provided and includes a container body defining a hollow interior, and a cap coupled to the container body to close off the hollow interior. The cap can include at least one recess having a figure-8 shaped projection with first and second openings therein. The first opening can include an inlet valve and the second opening can include an outlet valve, and the figure-8 shaped projection can be at least partially defined by first and second recesses formed in a surface of the cap.

(9) The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the substantially figure-8 shaped projection can include first and second collars defining the first and second openings and that are spaced a distance apart from one another, having the inlet and outlet valves disposed therein. For example, the first and second recesses surrounding the figure-8 shaped projection can each include first, second, and third sidewalls. The first and second sidewalls can be substantially convex and the third sidewall can be substantially concave. For example, the container body can have a substantially ovular cross-section with major and minor axes. The inlet and the outlet can be aligned along the minor axis. The cap can be configured to couple to the container body via a snap-fit.

(10) In one embodiment, a container for use in a beverage system is provided. The container includes a container body defining an interior hollow chamber and a cap covering the opening in the container body. The container body can have an opening leading to the interior hollow chamber. The cap can have an inlet port, an outlet port, and a collar positioned around the inlet port. The inlet port can have an inlet valve seated therein and can be movable between a closed configuration for preventing passage of fluid there through, and an open configuration for allowing passage of fluid

there through. The outlet port can have an outlet valve seated therein and movable between a closed configuration for preventing passage of fluid there through, and an open configuration for allowing passage of fluid there through. The collar can be positioned around the inlet port and can have an inner surface with at least a portion configured to circumferentially sealing engage a seal having an outer diameter in a range of about 7 mm to 8 mm.

(11) The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the body can include an end face having the inlet and outlet ports therein, and a skirt extending around the interface portion and defining a sidewall of the body. In some aspects, the skirt can have a substantially triangular shape. In other aspects, the collar can project outward from the end face. For example, the collar can be substantially cylindrical. For example, the inlet valve and the outlet valve each can include a cross-shaped slit configured to enable fluid flow therethrough. For example, the cap can include a closure pivotally coupled thereto and movable between an open position and a closed position. The closure can be configured to close off the inlet valve and the outlet valve in the closed position. In some aspects, the cap can include at least one closure retention feature on an external surface thereof, and the at least one closure retention feature can be configured to couple to the closure to retain the closure in the open position.

(12) In another embodiment, a container for use in a beverage system is provided. The container can include a container body defining an interior hollow chamber and a cap coupled to the opening of the container body. The cap can have an inlet valve that is sealed to retain the fluid additive within the interior hollow chamber and that is configured to open to allow gas to be injected into the interior hollow chamber, and an outlet valve that is sealed to retain the fluid additive within the interior hollow chamber and that is configured to open when a pressure within the interior hollow chamber exceeds a threshold pressure to allow fluid additive within the container body to flow through the outlet valve. The inlet valve can have a generally cylindrical shape and the outlet valve can have a generally cylindrical shape. A diameter of the outlet valve can be in a range from about 7 mm to 13 mm.

(13) The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the cap can include a closure pivotally coupled thereto and movable between an open position and a closed position. The closure can be configured to close off the inlet valve and the outlet valve in the closed position. For example, the container body can have a substantially ovular cross-section including a major axis about a first width and a minor axis about a second width. In some aspects, the inlet port and the outlet port can align with the minor axis of the container body. In other aspects, the cap can include at least one orientation element configured to orient the cap relative to the container body. For example, the inlet valve and the outlet valve each can include a cross-shaped slit configured to enable fluid flow therethrough.

(14) In one embodiment, a flow control assembly is provided. The flow control assembly can include a cap having a flow control system with an inlet port having an inlet valve and an outlet port having an outlet valve. The flow control system can achieve a Dosing Accuracy (DA) of about 100 or less according to the following formula:

$$(15) DA = \left[\frac{(Po - Pc)}{(Vd - Ls)} \right]$$

(16) Po is a pressure to open the outlet valve (mmH.sub.2O), Pc is a pressure to close the outlet valve (mmH.sub.2O), Vd is a diameter of the outlet valve (mm), and Ls is a length of the valve opening (mm).

(17) The flow control assembly can vary in a number of ways and may include any of the following features, alone or in combination. For example, the flow control system can achieve a DA of between about 40 and 70. For example, the flow control system can achieve a DA of about 55. For example, the pressure to open the inlet valve (Po) can be greater than about 100 mmH.sub.2O. For example, the pressure to open the inlet valve (Po) can be greater than about 400 mmH.sub.2O. For example, the diameter of the outlet valve (Vd) can be between about 5 mm and 15 mm, and in

certain embodiments can be about 9.5 mm. For example, the length of the valve opening (Ls) can be between about 1 mm and 5 mm, and in certain embodiments can be about 3.7 mm.

(18) In other embodiments, the cap can include a sidewall defining a cavity configured to receive a neck of a container. The cap can include an end wall having the inlet port and an outlet port formed therein. In some aspects, the inlet port and the outlet port each can include a cylindrical collar having the inlet valve and the outlet valve disposed therein, respectively. For example, the flow control assembly can include a container body defining an interior hollow chamber. The container body can have an opening leading to the interior hollow chamber, and the cap can be configured to couple to the opening of the container body to seal fluid within the interior hollow chamber. In some aspects, the inlet valve can be configured to allow a gas to be injected into the interior hollow chamber, and the outlet valve can be configured to open to allow fluid to flow out of the interior hollow chamber when a pressure within the interior hollow chamber exceeds the pressure to open the outlet valve (Po).

(19) In another embodiment, an ingredient container for use in a beverage carbonation system is provided. The ingredient container can include a container body defining an interior hollow chamber and an opening leading to the interior hollow chamber, and a cap coupled to the opening. The container body can have a cross-section with a major axis defining a width that is greater than a minor axis defining a depth. The cap can have an inlet that can be sealed to retain fluid within the container and that can be configured to open to allow gas to be injected into the interior hollow chamber. The cap can have an outlet that can be sealed to retain fluid within the container and that can be configured to open to allow fluid within the container to flow out through the outlet valve. The inlet and the outlet can be aligned along a first axis that extends parallel to the minor axis of the container body.

(20) The container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the first axis can extend substantially perpendicular to the major axis of the container body. For example, the cap can have an irregular shape. For example, the cap can have a substantially triangular outer perimeter. For example, the cap can have a major axis and a minor axis, and the first axis can extend along the minor axis of the cap. For example, the cross-section of the container body can be ovular.

(21) In another embodiment, an ingredient container is provided. The ingredient container can include a container body defining an interior hollow chamber and having an opening leading to the interior hollow chamber, and a cap positioned over the opening in the container body. The cap can have an irregular shape with a major axis and a minor axis, and the cap can include an inlet port and an outlet port positioned along the minor axis.

(22) The ingredient container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the inlet and outlet port can be positioned along an axis that extends substantially perpendicular to the major axis of the cap. For example, the cap can have a generally triangular cross-sectional shape. For example, the cap can have an outer perimeter with first, second, and third sides, and the first side can be longer than the second and third sides. In some aspects, the inlet and outlet valves can be positioned along an axis extending substantially perpendicular to the first side. For example, the cap can have a base wall having the inlet and outlet ports therein, and a sidewall extending around an outer perimeter of the base wall. The sidewall can have a height that varies around the outer perimeter. For example, the container body can have a cross-section with a major axis defining a width that is greater than a minor axis defining a depth, and the cap major axis can be aligned with the major axis of the container body.

(23) In another embodiment, an ingredient container is provided. The ingredient container can include a container body having a hollow interior and an opening leading into the hollow interior, and a cap positioned over the opening in the container body and including an inlet port and an outlet port. A cross-section of the cap can extend substantially perpendicular to a central axis of each of the inlet port and the outlet port can have a shape that is a substantially circular triangle.

(24) The ingredient container can vary in a number of ways and may include any of the following features, alone or in combination. For example, the cap can have an outer sidewall defining the shape of the cross-section and can have first, second, and third walls. In some aspects, the first wall can have a length that is greater than a length of each of the second and third walls. In other aspects, the first wall can be substantially planar, and the second and third walls can be convex. For example, the cap can have a base wall with the inlet and outlet ports therein, and an outer sidewall surrounding the base wall. The outer sidewall can have first and second shoulders projecting upward from the base wall. In some aspects, the cap can include a base wall having the inlet and outlet ports formed therein, and the base wall can include a circular cavity formed therein at a mid-portion thereof. In some variations, inlet and outlet ports can be positioned within the circular cavity.

(25) In another embodiment, an ingredient container is provided. The ingredient container can include a container body having an opening leading into a hollow interior, and a cap covering the opening. The cap can include a base having an inlet port and an outlet port formed therein, and a sidewall extending around the base and defining an outer perimeter of the cap body. The sidewall can include first and second shoulders extending upward from the base on opposed sides of the inlet and outlet ports. The first shoulder can have a first inner surface and the second shoulder can have a second inner surface. The first and second inner surfaces each can have a detent therein configured to receive a corresponding protrusion in a carriage assembly of a beverage carbonation system.

(26) The closure can vary in a number of ways. For example, the detent can include an opening formed through the first and second inner surfaces. In some aspects, the opening can be generally rectangular. For example, the closure can include a lid coupled to the cap body. The lid can be movable between an open position spaced a distance from the inlet and outlet, and a closed position in which the lid covers the inlet and outlet. For example, the sidewall can have a generally triangular cross-sectional shape. For example, the first and second inner surfaces can be substantially planar. For example, the first shoulder can have a first outer surface opposite the first inner surface, and the second shoulder can have a second outer surface opposite the second inner surface. The first and second outer surfaces can be convex. For example, the base can include a circular recess formed therein and can have the inlet and outlet port position therein.

(27) In another embodiment, a carbonation system is provided. The carbonation system can include a housing having at least one movable carriage with a cavity therein, and a container having a hollow body and a cap coupled to the hollow body. The cavity can include at least one spring-biased projection. The cap can include a base with inlet and outlet ports, and a sidewall extending around the base and having first and second shoulders, and at least one detent formed on an inner facing surface of at least one of the first and second shoulders. The at least one detent can be configured to receive the at least one projection in the carriage when the container is disposed within the cavity in the carriage.

(28) The carbonation system can vary in a number of ways. For example, the at least one projection and the at least one detent can be configured to produce an audible click when the container is inserted into the cavity in the carriage. For example, the at least one projection can include first and second projections positioned within the cavity, and the at least one detent can include first and second detents formed on the inner facing surface of the first and second shoulders, respectively. For example, the inner facing surface of the first and second shoulders can extend substantially perpendicular to the base. For example, the sidewall can have a substantially triangular cross-sectional shape.

(29) In another embodiment, a carbonation system is provided. The carbonation system can include a housing having at least one movable carriage with a cavity therein, and a container having a hollow body and a cap coupled to the hollow body. The cavity can include first and second spring-biased projections. The cap can include inlet and outlet ports, and the cap can have first and second

detents formed therein and configured to receive the first and second projections in the carriage when the container is disposed within the cavity in the carriage. The first and second projections and the first and second detents can be configured to produce an audible click when the container is inserted into the cavity in the carriage.

(30) The carbonation system can vary in a number of ways. For example, the cap can include a base having the inlet and outlet ports therein, and a sidewall can extend around the base and can include first and second shoulders. The first and second detents can be formed in the first and second shoulders, respectively. In some aspects, the first and second shoulders can have first and second inner facing surfaces with the first and second detents formed therein, and the first and second inner facing surfaces can extend substantially perpendicular to the base. For example, the cap can have a substantially triangular cross-sectional shape.

(31) In another embodiment, a container is provided. The container can include a container body having an opening extending into a hollow interior, and a cap extending across the opening. The cap can have an inlet port with an inlet valve configured to couple to a fluid source such that fluid can be delivered through the inlet valve to pressurize the hollow interior of the container body, and an outlet port with an outlet valve. The outlet valve can have a cracking pressure at which the outlet valve is configured to move from a closed configuration to an open configuration to dispense fluid from the hollow interior, and a closing pressure at which the outlet valve is configured to move from the open configuration to the closed configuration to prevent fluid from passing therethrough. The cracking pressure can be greater than the closing pressure.

(32) The container can vary in a number of ways. For example, a difference between the cracking pressure and the closing pressure can be in a range of about 300 mmH.sub.2O to 400 mmH.sub.2O. For example, a difference between the cracking pressure and the closing pressure can be about 340 mmH.sub.2O. For example, the cracking pressure can be greater than about 600 mmH.sub.2O or less than about 400 mmH.sub.2O. For example, the inlet valve and the outlet valve each can include a cross-shaped slit configured to enable fluid flow therethrough.

(33) In another embodiment, a container is provided. The container can include a container body having an opening extending into a hollow interior, and a cap extending across the opening. The cap can have an inlet port with an inlet valve configured to couple to a fluid source such that fluid can be delivered through the inlet valve to pressurize the hollow interior of the container body, and an outlet port with an outlet valve. The outlet valve can have a closed configuration to prevent fluid flow from the hollow interior, and can be movable to an open configuration to dispense fluid from the hollow interior in response to a pressure increase within the hollow interior increase of between about 300 and 380 mmH.sub.2O.

(34) The container can vary in a number of ways. For example, the pressure increase can be about 340 mmH.sub.2O. For example, the outlet valve can have a cracking pressure greater than about 600 mmH.sub.2O. The outlet valve can have a closing pressure less than about 400 mmH.sub.2O. In some embodiments, the inlet valve and the outlet valve can each have a cross-shaped slit configured to enable fluid flow therethrough.

(35) In another embodiment, a container is provided. The container can include a container body defining a hollow interior, and a cap. The cap can have an inlet port with an inlet valve seated therein and movable between a closed configuration for preventing passage of fluid there through, and an open configuration for allowing passage of fluid there through. The cap can also have an outlet port having an outlet valve seated therein and movable between a closed configuration for preventing passage of fluid there through, and an open configuration for allowing passage of fluid there through. The outlet valve can have a configuration that will dispense a predetermined amount of fluid in a range of 1.6 mL to 2.0 mL in response to a dose of gas being pumped into the container for a period of 140 ms.

(36) The container can vary in a number of ways. For example, the predetermined amount of fluid can be 1.8 mL. For example, the inlet valve and the outlet valve each can include a cross-shaped

slit configured to enable fluid flow therethrough. For example, the outlet valve can have a cracking pressure at which the outlet valve is configured to move from a closed configuration to an open configuration to dispense fluid from the hollow interior, and can have a closing pressure at which the outlet valve is configured to move from the open configuration to the closed configuration to prevent fluid from passing therethrough. The cracking pressure can be greater than the closing pressure. For example, the predetermined amount of fluid is proportional to a difference between the cracking pressure and the closing pressure.

(37) The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

(1) These and other features will be more readily understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a front view of one embodiment of a beverage dispensing system;

(3) FIG. 2 is a rear perspective view of the beverage dispensing system of FIG. 1 with various housing components removed;

(4) FIG. 3 is a front perspective view of one embodiment of a housing portion and carriage assembly for use with a beverage dispensing system;

(5) FIG. 4 is a perspective view of a carriage assembly of FIG. 3 having the housing portion removed;

(6) FIG. 5A is a top view of a carriage used with the carriage assembly of FIG. 3;

(7) FIG. 5B is a cross-sectional view of the carriage of FIG. 5A;

(8) FIG. 6 is a bottom view of the carriage of FIG. 5A;

(9) FIG. 7 is a perspective view of an ingredient container according to an embodiment;

(10) FIG. 8A is a cross-sectional view of the ingredient container of FIG. 7;

(11) FIG. 8B is a partial cross-sectional view of the ingredient container of FIG. 7;

(12) FIG. 9 is an exploded view of the ingredient container of FIG. 7;

(13) FIG. 10A is a perspective view off a container body of the ingredient container of FIG. 7;

(14) FIG. 10B is a top view of the container body of FIG. 10A;

(15) FIG. 11A is a perspective view of a lid of the ingredient container of FIG. 7;

(16) FIG. 11B is a top view of the lid of FIG. 11A;

(17) FIG. 12A is a perspective view of an outlet valve of the ingredient container of FIG. 7;

(18) FIG. 12B is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

(19) FIG. 12C is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

(20) FIG. 12D is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

(21) FIG. 12E is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

(22) FIG. 12F is a cross-sectional view of the outlet valve of FIG. 12A during a dispensing process;

(23) FIG. 13 is a rear perspective view of the lid of FIG. 11A;

(24) FIG. 14 is a perspective cross-sectional view of the lid of FIG. 11A;

(25) FIG. 15 is a partial perspective rear view of the lid of FIG. 11A having a lid cover in a closed position;

(26) FIG. 16 is a perspective bottom view of the lid of FIG. 11A;

(27) FIG. 17 is a perspective view of the carriage assembly of FIG. 3 having the ingredient container of FIG. 7 loaded therein;

(28) FIG. 18 is a perspective view of the carriage assembly and ingredient container of FIG. 17

having a housing removed;

(29) FIG. 19 is a perspective view of the carriage of FIG. 5A having the ingredient container of FIG. 7 loaded therein;

(30) FIG. 20 a cross-sectional view of the carriage and ingredient container of FIG. 19;

(31) FIG. 21 is a partial cross-sectional perspective view of the container and ingredient container of FIG. 19; and

(32) FIG. 22 is a bottom view of a carriage assembly according to some embodiments showing a relative position of ingredient container outlets and a fluid outlet in relation to variously-sized drinkware.

(33) It is noted that the drawings are not necessarily to scale. The drawings are intended to depict only typical aspects of the subject matter disclosed herein, and therefore should not be considered as limiting the scope of the disclosure.

DETAILED DESCRIPTION

(34) Certain illustrative embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting illustrative embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one illustrative embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.

(35) Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape.

(36) In general, ingredient containers for use with beverage dispensers and carriages for receiving ingredient containers are provided. In one embodiment, an ingredient container is provided that can contain an additive for use in a beverage dispensing process. The ingredient container can have a hollow container body with an opening and a lid coupled to the container body. The lid can include a lid base configured to couple to the container body over the opening, and the lid base can have an inlet and an outlet therein. In certain embodiments, the lid can further include a lid cover configured to selectively close the inlet and the outlet, thereby sealing a hollow interior of the container body. The inlet and the outlet can each have a seal disposed therein that is configured to open in the presence of a pressure differential between an interior and an exterior of the ingredient container in an attempt to eliminate the pressure differential. The ingredient container can be shaped and designed to correspond to a carriage located on a beverage dispensing device. The carriage can have complimentary features to receive and retain the ingredient container, and when retained, the ingredient container can be employed by a beverage dispensing device for use in the creation of customized beverages.

(37) Methods of dispensing the additive stored within the ingredient container can vary. In some embodiments, the ingredient container is pressurized with a gas, such as air, to cause the outlet to open and dispense the stored additive. When the ingredient container is properly seated and retained by a carriage, a gas line fluidly coupled to a pump can receive the inlet of the ingredient container in order to seal around the inlet in preparation for the introduction of gas into the ingredient container during a dispensing procedure. Gas can be pumped by the pump, through the gas line, through the inlet seal, and into the hollow interior of the ingredient container. The

resulting increase in internal pressure can cause the outlet seal to open and dispense an amount of the additive proportional to the amount of gas introduced through the inlet.

(38) FIGS. **1-2** illustrate a beverage dispensing system **10** according to one embodiment. The beverage dispensing system **10** can be used to create and dispense customized beverages for a user, based on desired characteristics of the beverage. The illustrated beverage dispensing system **10** generally includes a housing **12** having a fluid reservoir **14** and a carbonation assembly **16**. A carriage assembly **18** can be included on and/or coupled to the beverage dispensing system **10**, and it can receive one or more ingredient containers **20** to be used in the creation of beverages. The ingredient containers **20** can include one or more additives (e.g., a flavorant, a vitamin, a food dye, etc.) to be included in a created beverage as desired.

(39) During a beverage dispensing process, a user can actuate inputs located at a user interface **22** in order to select specific characteristics of the desired beverage, such as volume, carbonation level, specific additives, and additive amount. If the user selects inputs to indicate that the beverage is carbonated, water can be fed from the fluid reservoir **14** and into the carbonation assembly **16**, and carbon-dioxide can be fed from a canister **24** and into the carbonation assembly **16** to produce carbonated water. If the user selects inputs to indicate that one or more additives should be added to the beverage, the beverage dispensing system **10** can dispense the additive from the one or more ingredient containers **20** coupled to the system. The beverage can be dispensed into a container, such as a drinking glass **26**.

(40) FIGS. **3-6** illustrate one embodiment of a carriage assembly **100** which can be coupled to and/or retained with or within a beverage dispensing device, such as beverage dispensing device **10**. In the illustrated embodiment, the carriage assembly **100** is contained within a carriage housing **100A**. The carriage assembly **100** can include one or more carriages **101**, which can each seat and retain one or more ingredient containers (not shown) for use in a beverage dispensing process. Although the carriage assembly **100** is shown having two separately movable carriages **120**, a different number of carriages **120** are contemplated herein as well. For example, the carriage assembly can be in the form of a single movable carriage having multiple cavities with each cavity configured to receive an ingredient container. Ingredient containers and their retention within the carriage assembly **100** will be described in greater detail below.

(41) FIG. **4** illustrates the carriage assembly **100** separated from the carriage housing **100A**. The illustrated carriage assembly **100** generally includes a left carriage **120L** and a right carriage **120R** (collectively carriages **120**) coupled to a carriage base **110**. The carriage base **110** can have a variety of forms, which may depend upon the form of the carriage housing **100A** containing the carriage base **110**. As illustrated, the carriage housing **100A** and the carriage base **110** have a substantially cylindrical form. The carriage base **110** can include cutouts and/or slots for seating and receiving various components including, for example, the carriages **120** and a fluid outlet **114**. The carriages **120** can be coupled to the carriage base **110** in a variety of ways, for example, the carriages **120** can be pivotally hinged to the carriage base **110** such that the carriages can pivot downward in order to facilitate loading one or more ingredient containers. The left carriage **120L** is illustrated in FIGS. **3** and **4** in an upward position, while the right carriage **120R** is pivoted downward to a downward position. The carriages **120** can be coupled to the carriage assembly **100** in other ways, such as via a sliding connection, a stationary connection, etc., or they can be coupled directly to a beverage dispensing device. The illustrated carriage base **110** further includes lift assists **116**, which can be coupled to a rear region of the carriages **120**. The lift assists **116** can include a biasing feature such as a spring, such that each of the coupled carriages are biased to the upward position. A micro-switch **112** (also referred to as left micro-switch **112L** and right micro-switch **112R**) can be located above each of the carriages **110**, which will be discussed in more detail below.

(42) FIGS. **5** and **6** depict a single carriage **120** in more detail. Features described as applying to one carriage can be applied to all carriages. As shown, the carriage **120** has a generally rectangular

carriage body **122** with a rounded front face **124** that can be shaped to conform with an overall contour of the carriage housing **110A**. A handle **128** can extend from the front face **124** to provide a grasping surface to enable the carriage **120** to be easily pivoted, such as when an ingredient container is placed into or removed from the carriage assembly **120**. While the handle **128** is shown in the form of a protruding lip or ledge, the handle **128** can take on various forms and can include protrusions of other shapes as well as recesses within the carriage body **122** itself. The carriage **120** can further include a pivot axis **126** located near a rear of the carriage body **122**, as introduced above, for allowing the carriage **120** to pivot relative to the carriage body **122**.

(43) An upper portion of the carriage body **122** can include a carriage face **130**, as best shown in FIGS. 5A and 5B. In an exemplary embodiment, the carriage face **130** is shaped to receive and retain a complimentary ingredient container for use during a beverage dispensing process. The carriage face **130** can include a variety of indentations, protrusions, flat areas, and rounded areas to fully receive ingredient containers of any shape or size, as well as to ensure that an ingredient container is properly seated and coupled to the system. In the illustrated embodiment, the carriage face **130** is in the form of a generally triangular recess with rounded corners, e.g., a rounded triangle. One side can include a rectangular cutout **131** extending therefrom. In certain embodiments, the cutout **131** can be formed in the longed side of the triangle, and it can be located closest to a mid-portion of the carriage assembly **100**. The central region **132** of the carriage face **130** can include a raised platform having a variety of features thereon. As shown, the central region **132** is raised such that a peripheral channel **133** is defined within the carriage face **130**. The central region **132** can include a carriage inlet **134** and an carriage outlet **136**, which can be configured to align with and couple to an inlet and an outlet of an ingredient container, respectively. The illustrated carriage inlet **134** and the carriage outlet **136** have a substantially round form defining a central opening **134A**, **136A**. The central openings **134A**, **136A** can pass entirely through the carriage **120**. The inlet and outlet receivers **134**, **136** can be made from a variety of materials. For example, one or both of the inlet and outlet receivers **134**, **136** can be made from a plastic, a resin, a rubber, a metal, or a composite thereof. In certain embodiments, for example, one or more of the inlet and outlet receivers **134**, **136** can be made from a rubber or rubber-like material such that an air-tight seal is created between the carriage face and a seated ingredient container, as discussed further below.

(44) A space around the carriage inlet **134** and the carriage outlet **136** can be recessed into the central region **132**, thereby defining the overall form of both the carriage inlet **134** and the carriage outlet **136**. In the illustrated embodiment, this space, also called the central recess **137**, takes the form of a substantially “figure-8” shape, with the inlet and outlet receivers **134**, **136** being positioned within each opening of the “figure-8.” The central region **132** can also include one or more flanking protrusions **138** disposed proximate to the central recess **137**. The flanking protrusions **138** can be informed by the shape of other features found in the carriage face **130**, or they can have independent designs. In the illustrated embodiment, the flanking protrusions **138** are shaped to extend into complimentary recesses on an ingredient container to assist in the retention thereof. As shown in FIGS. 5A and 5B, the carriage face **130** includes a pair of similar flanking protrusions **138**, which take the form of “bat wings” that follow the “figure-8” contour of the central recess **137**. In particular, the illustrated flanking protrusions **138** have an outer sidewall that is convexly curved along its length and two inner sidewalls that are each convexly curved to follow the contours of the inlet and outlet receivers **134**, **136**. The sidewalls of the flanking protrusions **138** can taper in a direction leading away from the carriage face **130**, as shown, such that the tip portion is generally smaller in size than the base portion of each protrusion **138**. In other embodiments, the protrusions **138** may not flare at all.

(45) The peripheral channel **133** can further include one or more features to assist in the retention of an ingredient container. As will be discussed in more detail below, each peripheral channel **133** can have a shape configured to complement a shape of the container such that two shoulders on the

container, as well as other portions of the container, can be received therein. In the illustrated embodiment, the peripheral channel **133** includes two generally rounded triangular areas and an elongated slot extending therebetween. The channel **133** is defined by the shape of the center region **132**, which is generally square with rounded corners, in combination with the shape of the generally triangular recess in the carriage face **130**. The peripheral channel **133** can also include one or more retainers **139** protruding from a sidewall of the center region **132** outward into the peripheral channel **133**. The retainers **139** can be spring-biased outward, such that during a retention process the retainers **139** can be forced inward by the container before springing back outward to engage a corresponding recess in the ingredient container. The retention process will be described in greater detail below.

(46) FIG. 5B illustrates the relative heights of the carriage **120**, including the carriage inlet **134**, the carriage outlet **136**, and the flanking protrusions **138**. As shown, the carriage outlet **136** has a height that is greater than a height of the carriage inlet **134**. The retainers **139** can be seen located within the peripheral channel **133**, which is set below the elevated center region **132**. While the carriage face **130** is described and shown as having certain areas recessed and other areas protruding, carriage faces with the opposite features are contemplated herein as well, i.e., all protrusions are recesses and all recesses are protrusions. Further, carriage faces are also contemplated that may have only a portion of the features interchanged, such that only one or a few protrusions are recesses and/or only one or a few recesses are protrusions. Other shapes and configurations are also contemplated.

(47) FIG. 6 depicts an underside of the carriage **120**, according to some embodiments. The underside of the carriage **120** is positioned on the opposite side of the carriage inlet **134** and the carriage outlet **136**, and it includes central holes **134A**, **136A**, which, as introduced above, can pass through the carriage **120**. In operation, the central hole **134A** of the carriage outlet **136** can be coupled to a gas line **140**. The gas line **140** can be coupled at an opposite end to an air pump (not shown), which can be used to introduce air or another gas into a seated ingredient container. The resulting increase in pressure can cause the seated ingredient container to dispense a stored additive through the central hole **136A** of the carriage outlet **136**. In systems with more than one carriage, one or more pumps can be used to introduce gas to a seated ingredient container. In some variations, each carriage can have its own pump fluidly coupled thereto via a gas line or similar setup. In other embodiments, the gas line **140** can be coupled to the carbonation source, which can be used to supply gas to the container for ejecting additive.

(48) FIGS. 7-16 illustrate an exemplary embodiment of an ingredient container **200**. The ingredient container **200** can generally include a lid **210** coupled to a container body **250** which can be configured to contain an additive (e.g., a flavorant, a supplement, a vitamin, a coloring agent, etc.) to be used in the creation of beverages. The additive can be in the form of a fluid, a solid, a powder, a gel, a syrup, or any other form. The ingredient container **200** can come in a variety of sizes. For example, the ingredient container **200** can have an overall height between about 55 mm and 60 mm, and in some embodiments can be about 56.9 mm. The ingredient container **200** can have a maximum width between about 55 mm and 65 mm, and in some embodiments, the maximum width can be about 59.5 mm. The lid **210** can have a depth between about 38 mm and 42 mm, and in some embodiments can be about 39.6 mm. The container body **250** can have a depth between about 38 mm and 42 mm, and in some embodiments can be about 39.5 mm. For example, the ingredient container **200** can have a volume between about 50-90 mL, and in some variations can have a volume of about 70 mL.

(49) The ingredient container **200** can store the additive inside, and, as part of a beverage creation process, receive a measured volume of gas (e.g., air, carbon-dioxide, etc.) through an inlet **224** resulting in an increased internal pressure. The increase in internal pressure within the container **200** can result in an outlet **226** emitting a tailored amount of the additive as a consequence of eliminating or reducing the newly-created pressure differential across the outlet.

(50) The illustrated container body **250** has a generally oblong, ovular form similar to a race-track configuration, as seen in FIGS. **10A** and **10B**. While the container body **250** is shown as having a specific form, the container body **250** can take on a variety of forms. This oblong ovular form can include a minor width **W1** about a shorter dimension of the container body **250** and a major width **W2** about a longer dimension of the container body **250**. Similarly, the oblong, ovular form can have a minor axis **A1** extending centrally along the minor width **W1**, and the oblong, ovular form can have a major axis **A2** extending centrally along the major width **W2**. As will be discussed in more detail below, the shape of the container body can aid in allowing multiple containers to be positioned closer to one another within the beverage system, thus allowing the outlets **226** to be positioned closer for dispensing an additive.

(51) The container body **250** can include a base **252**, a sidewall **254** extending upwardly from the base **252**, and a top **256**, which together can define an interior space to store the additive. In some embodiments, the base **252** can include an ovular recess **253** as shown in FIG. **8B**. The ovular recess **253** can provide increased structural integrity to the container body **250** during storage, transit, operations, etc., and it can also provide an area for increased engagement, such as by a user and/or by a beverage dispensing device (e.g., beverage dispensing system **10**).

(52) The sidewall **254** can extend upward from the base **252** to maintain a substantially constant cross-section. The sidewall **254** can include first and second side faces **254A**, **254B**, which can be substantially planar, and first and second convexly curved faces **254C**, **254D** extending between the first and second side faces **254A**, **254B**. A series of channels **255** can run vertically on the first and second side faces **254A**, **254B**, substantially parallel to each other. The channels **255** can operate similarly to the ovular recess **253**, in that they may provide for increased structural integrity, and/or they may provide an area of increased engagement between the container body **250** and a beverage dispensing device (e.g., beverage dispensing system **10**). They can also aid in gripping the container. In certain embodiments, a carriage assembly (e.g., carriage assembly **100**) can have complimentary components to be received by the channels **255** in order to aid in retention of the ingredient container **200**.

(53) The top **256** sits upon the sidewall **254**, and it can include a shoulder **258** and a neck **260**. The shoulder **258** can have a gradual slope upward toward the neck **260**, which can be centrally disposed on the top **256** and can be a round, substantially vertical portion of the container body **250**. The neck **260** can define the opening **262** leading to the interior of the container body **250**. A circumferential flange **264** can extend around the neck **260** and can provide a coupling point for the lid **210**, such as with a snap-fit. In some embodiments, the circumferential flange **264** can be replaced by threads to provide threaded connection with the lid **210**. A pair of orientation protrusions **266** can be disposed on opposite sides of the neck **260**. These protrusions **266** can vary in shape or number, and they can function to align with complimentary features on the lid **210** to ensure that the lid **210** is properly oriented on the container body **250**.

(54) FIGS. **11A-16** depict the lid **210** and elements thereof, separated from the container body **250**. The illustrated lid **210** has a substantially rounded triangular shape and includes a lid base **220** and a lid cover **240** coupled to the lid base **220**. The lid cover **240** can be used to close the inlet and outlet, and in turn the container body **250**. The triangular shape can be defined by a perimeter having first, second, and third sides, with the first side being longer than each of the second and third sides.

(55) The lid base **220** can include a skirt **222** located at a lower perimeter thereof and having a curved shaped to conform with the shoulder **258** of the container body **250**. The skirt **222** can include a front recess **223**, which can be shaped to allow a portion of the lid cover **240** to extend outward beyond the skirt **222** when the lid **210** is in the closed position to enable grasping of the lid cover **240** to case opening and closing of the lid cover **240** relative to the lid base **220**. The lid base **220** can include an inlet **224** and an outlet **226**, which lead respectively to and from the interior of the container body **250**. The inlet **224** can include an inlet collar **224A** flanking an inlet orifice

224B, while the outlet **226** can include an outlet collar **224A** flanking an outlet orifice **224B**. In the illustrated embodiment, the inlet collar **224** has a height that is greater than a height of the outlet collar **226**. The greater height of the inlet collar **224** can aid in allowing a seal to be formed between the container inlet **224** and the outlet **136** on the carriage **120**.

(56) The container inlet **224** and the outlet **226** can be positioned on the lid base **220** in line with a minor axis B-B of the lid **210** extending along a plane defined by an upper face of the lid base **220**, as shown in FIG. **11A**. When the lid **210** is coupled to the container body **250**, the minor axis B-B can extend parallel to the minor axis A1 of the container body **250**, and therefore can extend perpendicular to the major axis A2. In some variations, the entire lid **210** can be substantially symmetrically mirrored about the minor axis B-B. The lid **210** can also have a major axis A-A, as seen at least in FIGS. **11A-11B**, which can extend perpendicular to the minor axis B-B.

(57) As shown in FIG. **11B**, the inlet **224** and outlet **226** can each have a central longitudinal axis (also called a central axis) with a distance D there between. The central longitudinal axis of each of the inlet **224** and the outlet **226** is coming out of the page in FIG. **11B**, but it is shown from a side view in FIG. **21**. The distance D between each central longitudinal axis can vary. In certain embodiments, the distance D can depend at least partially on the overall dimensions of the lid **210** and/or the sizes of the valves, as discussed further below. For example, in some embodiments, the distance D between the central axes can be between about 9 mm and 15 mm, and more preferably between about 11 mm and 13 mm, and in certain exemplary embodiments the distance D can be about 13 mm.

(58) As further shown in FIG. **11B**, the inlet **224** can have a diameter X1, and the outlet **226** can have a diameter X2. The inlet diameter X1 can be between about 6.6 mm and 7.2 mm, and in some embodiments can be about 6.90 mm. The outlet diameter X2 can be between about 6.5 mm and 7.1 mm, and in some embodiments can be about 6.84 mm.

(59) Recesses **228** can flank each side of the inlet **224** and the outlet **226**, and the recesses **228** can each be shaped to correspond to protrusions in a carriage (e.g., flanking protrusions **138** on carriage **120**). For example, the recesses **228** can be shaped to follow an outer contour of the collars **224A**, **226A** and can take a “bat wing” form. In particular, similar to the flanking protrusions **138**, the recesses **228** can have a radially outward sidewall that is concavely curved along its length and two inner sidewalls that are concavely curved to follow the contours of the inlet and outlet **224**, **266**. The recesses **228** can take on various other forms as well, and their form may be at least partially dependent upon the placement and form of other components on the lid **210**. The recesses **228** can be placed a slight distance apart from the inlet **224** and the outlet **226**, thus defining a central pattern **230** located in the space between the collars **224A**, **226A** and the recesses **228**. As best seen in FIG. **11A**, the central pattern **230** can take the form of a “figure-8,” however other forms may be present. The illustrated central pattern **230** is shown being flush with the upper surface of the base **220**, however the central pattern **230** can protrude above the upper surface or can be recessed below the upper surface. The central pattern **230** can be a protrusion, a recession, or a combination thereof with a portion of the central pattern **230** protruding from the lid **210** and a portion of the central pattern **230** receding into the lid **210**. In some variations, the inlet and outlet collars **224A**, **226A** can contribute to the central pattern **230**.

(60) As explained previously with respect to the carriage face **130**, although the lid base **220** is described and shown as having certain areas recessed and other areas protruding, lid bases with the opposite features are contemplated herein as well, i.e., all protrusions are recesses and all recesses are protrusions. Further, lid bases are also contemplated that may have only a portion of the features interchanged, such that only one or a few protrusions are recesses and/or only one or a few recesses are protrusions.

(61) The lid base **220** can further include a pair of shoulders **231** formed on opposed sides of the skirt **222** and that extend upward from the lid base **220**. Each shoulder **231** can have a shape, such as a rounded triangular shape, that complements a shape of the peripheral channel **133**. Each

shoulder **231** can also include one or more retention features, which can further assist in retention of the ingredient container **200** within the carriage **120**. These features can be in the form of receivers **232** which can receive a complimentary element of the carriage **120**, as will be described in more detail below. In the illustrated embodiment, the receivers **232** are each in the form of a substantially square or rectangular recess or cut-out formed in an inward facing sidewall of each shoulder **231**.

(62) As further shown, a rear portion of the lid base **220** can include a rear wall **233** which can extend between the shoulders **231**. The lid cover **240** can be coupled to the rear wall **233**, as will be discussed in more detail below.

(63) Referring again to the inlet **224** and the outlet **226**, as previously explained the inlet **224** can include an upwardly extending inlet collar **224A** flanking an inlet orifice **224B**, and the outlet **226** can include an upwardly extending outlet collar **226A** flanking an outlet orifice **226B**. Although the inlet collar **224A** and the outlet collar **226A** are shown in a circular form, the inlet and outlet collars **224A**, **226A** can take on a number of shapes, including various geometric shapes, e.g., a triangle, a star, etc., as well as fanciful and/or irregular shapes, e.g., a letter, a logo, etc. The form of the inlet and outlet collars **224A**, **226A** can be the same or different. As shown in FIG. **9**, the inlet **224** can include an inlet valve frame **224C** and an inlet valve **224D**, and the outlet **226** can include an outlet valve frame **226C** and an outlet valve **226D**. Generally, each of the inlet valve **224D** and the outlet valve **224D** can be respectively seated within the inlet valve frame **226C** and the outlet valve frame **226C**. The inlet valve frame **224C** and the outlet valve frame **226C** can be affixed to the underside of the lid **210** beneath the inlet **224** and the outlet **226** respectively. In other embodiments, the inlet valve frame **224C** and the outlet valve frame **226C** can be formed from a single frame component.

(64) FIG. **12A** depicts one embodiment of an outlet valve **226D** in more detail. While description is made with respect to the outlet valve **226D**, similar features are applicable to the inlet valve **224D**. Additionally, where options are provided for aspects of the outlet valve **226D**, actual aspects may not always be the same between the inlet valve **224D** and the outlet valve **226D**. The illustrated outlet valve **226D** is configured to open to dispense an additive therefrom during a beverage dispensing process. While the outlet valve **226D** is depicted as being round or substantially circular, the outlet valve **226D** can vary in form to have any number of regular or irregular shapes. In general, the outlet valve **226D** can include a flange **226E** configured to hold an outlet valve head **226F** within the outlet valve frame **226C**. The flange **226E** can be connected to the outlet valve head **226F** via a roll sleeve **226G**. The outlet valve **226D** can also vary in size, and the size can depend at least in part on the diameter of the outlet **226** itself. For example, the outlet valve diameter V_d of the outlet valve **226D** on the container body **250**, i.e., not including the flange **226E**, can be between about 8 mm to 12 mm. In some embodiments, the outlet valve diameter V_d can be between about 9 mm and 10 mm. The outlet valve **226D** can be in the form of a slit valve having a slit **226H** configured to open and allow for the transfer of a material, such as a fluid, therethrough. The slit **226H** can have a variety of forms and sizes. For example, as shown in FIG. **12A**, the slit **226H** has a cross or X shape. The slit **226H** can vary in size, but in an exemplary embodiment it can have a slit length L_s between about 1.5 mm and 5.5 mm. Note the slit length as used herein refers to the length of the longest slit where two or more slits are provided. In some embodiments, the slit length L_s can be between about 1.5 mm and 2 mm, and the outlet valve **226D** can open at the cross-shaped slit **226H** when subjected to enough pressure, either internally or externally. An opening pressure P_o (also called a cracking pressure) of the outlet valve **226D** can vary, and can be dependent upon the material, size, or other details of the outlet valve **226D**. For example, in some embodiments, the opening pressure P_o can be about 300 mmH.sub.2O or greater, and more preferably about 600 mmH.sub.2O or greater. A closing pressure P_c of the outlet valve **226D** can vary as well, and can be dependent upon various details of the outlet valve **226D**. In some embodiments, the closing pressure P_c can be about 400 mmH.sub.2O or less. In other embodiments, the closing pressure P_c can be about 300 to 400 mmH.sub.2O less than the cracking

pressure.

(65) When the outlet valve **226D** is subjected to a high enough pressure differential, such as in the build-up to attaining the opening pressure P_o and then subsequent achievement of the opening pressure P_o , the valve **226D** can undergo a several-step transformation process before opening at the slit **226H**. This transformation process is illustrated in FIGS. **12B-12F**. In FIG. **12B**, the outlet valve head **226F** begins to move downward, subject to some pressure, rolling about the outlet valve sleeve **226G**. In FIG. **12C**, the outlet valve sleeve **226G** is fully unrolled. In FIG. **12D**, the outlet valve head **226F** begins to flatten, and then at FIG. **12E**, the opening pressure P_o is achieved, forcing the slit **226H** open and dispensing an additive. When the slit opens, the pressure differential across the valve **226D** quickly dissipates, and the valve head **226F** can return to its typical position. As a result of this return, in some configurations, the slit **226H** can open inwardly, as shown in FIG. **12F**, before finally reaching a rest state and returning to the position depicted in FIG. **12A**. In other configurations, an internal pressure on the outlet valve **226D** can cause the outlet valve **226D**, after opening, to return to the state depicted in FIG. **12B**, and the outlet valve **226D** may never fully return to the state shown in FIG. **12A**.

(66) In some embodiments, the inlet valve **224D** can be positioned in the same orientation as the outlet valve **226D**. In these embodiments, fluid flows through the inlet valve **224D** in the opposite direction as the fluid flowing through the outlet valve **226D**, i.e., fluid flows into the ingredient container **200** through the inlet valve **224D** but flows out of the ingredient container **200** through the outlet valve **226D**, all while the inlet and the outlet valves **224D**, **226D** are positioned in the exact same orientation. As a result, in these embodiments, the inlet valve **224D** does not undergo the same series of steps shown in FIGS. **12A-12F** when fluid flows therethrough. Instead, the inlet valve begins in the state shown in FIG. **12A** and when subjected to pressure great enough to open the inlet valve **224D**, the inlet valve **224D** merely opens in a manner similar to the state shown in FIG. **12F**, but facing the direction shown in FIG. **12A**, thus allowing fluid to flow through the opening. Because fluid is flowing through the inlet valve **224D** in a direction that is opposite a direction of fluid flowing through the outlet valve **226D**, the inlet valve **224D** does not undergo the series of steps involving rolling to an expanded state and then opening, as depicted in FIGS. **12B-12E**.

(67) As previously indicated, the lid **210** can also include a lid cover **240**, shown in FIGS. **11** and **13**, which can be connected to the rear wall **233** by various means, including by a hinge **234** (e.g., a living hinge). The lid cover **240** can include an inlet cover **242** and an outlet cover **244**, which are sized to respectively close the inlet **224** and the outlet **226** on the lid base **220**. Each of the inlet cover **242** and the outlet cover **244** can include respective inlet and outlet cover collars **242A**, **244A** that are sized to be internally received by the inlet collar **224B** and the outlet collar **226B**, as seen in the cross-section of FIG. **14**. The outlet cover **244** can also include a central plug **244B** that is sized to be internally received by the outlet **226** itself. The central plug **244B** can operate to prevent premature opening of the outlet valve **226A**. The central plug **244B** can protrude out from the lid cover **240** beyond the protrusion distance of the outlet cover collar **244A** in order to facilitate closure of the outlet **242** when the lid cover **240** is in the closed position.

(68) In some embodiments, the lid can include features to hold the cover **240** in an open position. For example, as shown in FIG. **15**, the lid base **220** can include a back side **235** having a substantially flat central face **236** with a width that is substantially equal to a width of the lid cover **240**. One or more lid cover retention features **236** can be located at an upper end of the back side **235** near the hinge **234**. These features **236**, which can be in the form of cut-outs or recesses, can secure the lid cover **240** when the lid cover **240** is in an open position. As shown, the lid cover **240** can include cover tabs **246** extending from at least one side of the lid cover **240**. The cover tabs **246** can extend into the cover retention features **236** to assist in retention of the lid cover **240** in the open position. In the closed position, the inlet and outlet cover collars **242A**, **244A**, as explained previously, can extend into and frictionally engage the inlet **224** and the outlet **226**. This frictional

engagement can assist in retention of the lid cover **240** in the closed position. Additionally, the inlet and outlet cover collars **242A** can prevent the inlet and outlet valves **224D**, **226D** from opening prematurely, such as during transportation. For example, the outlet valve **226D** can be prevented from rolling about the roll sleeve **226G** as illustrated in FIGS. **12B-12D**.

(69) FIG. **16** depicts an underside of the lid **210**. The lid base **220** can have a divided, arcuate rim **229** that is sized to couple with the neck **260** (not shown) of the container body **250**. The arcuate rim **229**, can couple to the neck **260** (not shown) of the container body **250** (not shown) via a snap-fit, threads, and the like. For example, the arcuate rim **229** can include an internal lip **229A** that is configured to interface with the flange **264** located on the neck **260**. This engagement can be seen especially in FIG. **8C**. In an exemplary embodiment, the arcuate rim **229** includes a ridge that engages a corresponding feature on the neck **260** to form a snap-fit connection. Depending upon the means by which the arcuate rim **229** affixes to the container body **250**, the physical structure of the arcuate rim **229** may change accordingly. While not shown, a seal such as an O-ring can be disposed within the rim **229** to aid in coupling the lid **210** to the container body **250**.

(70) As further shown in FIG. **16**, an inner surface of the skirt **221** can include one or more orientation channels **227** that can receive the orientation protrusions **266** (not shown) found on the top **256** of the container body **250** to aid in orientation of the lid **210** on the container body **250**. As a result, the lid **210** can be limited to mating to the container body **250** in only two orientations. In embodiments where the lid **210** includes one or more recesses **228** to facilitate coupling with a carriage (e.g., carriage **120**), the recesses **228** can extend downward from the underside of the lid **210** in between the inlet and outlet valve frames **224C**, **226C**, and the arcuate rim **229**. In some variations where the recesses **228** are at least partially defined by the shape of the inlet **224**, the outlet **226**, and the overall shape of the lid **210**, the recesses **228** can occupy the entirety of the space found between the inlet and outlet valve frames **224C**, **226C**, and the arcuate rim **229**. Essentially, the recesses **228** can change in form depending upon other features located on the lid **210**, such as the inlet collar **224A**, the outlet collar **226A**, the inlet valve frame **224C**, the outlet valve frame **226C**, the arcuate rim **229**, the retention pattern **230**, and more. In other variations, the recesses **228** can occupy only a portion of this space.

(71) FIGS. **17-22** depict the ingredient container **200** retained within the carriage assembly **100**. FIG. **18** depicts the carriage assembly **100** with the carriage housing **110A** (not shown) and the right carriage **120R** (not shown) removed. With the right carriage **120R** removed, the fluid outlet **114** is more visible.

(72) FIGS. **19-22** depict the ingredient container **200** seated within a carriage **120** in greater detail and from various angles to illustrate components on the ingredient container **200** and the carriage **120** coupling together. In order to seat the ingredient container **200**, the lid cover **240** can be retained in the open position, as explained above. The carriage **120** can be lowered to expose the carriage face **130**, and the ingredient container **200** can be aligned with the carriage face **130** and pressed down so that the carriage inlet **134** engages the inlet **224** and the carriage outlet **136** engages the outlet **226**. When seated, the flanking protrusions **138** can extend into the recesses **228**, as best shown in FIG. **20**. Both the carriage inlet **134** and the carriage outlet **136** can extend respectively into the inlet **224** and the outlet **226**, and the inlet collar **224A** and the outlet collar **226A** can extend circumferentially around the carriage inlet **134** and the carriage outlet **136**, as best shown in FIGS. **21** and **22**.

(73) While not shown in FIGS. **19-22**, in embodiments with a retainer **139**, the retainer **139** can also snap into the receivers **232**, which may provide an audible signal for a user to know that engagement is successful, such as an audible click when the retainer **139** is engaged and no longer under tension. The rectangular cutout **131** in the carriage face **130** can receive the lid cover **240** when the lid cover **240** is in the fully-opened position, as seen in FIG. **19**. Once the container **200** is fully seated, the carriage **120** can be returned to an elevated position. In some embodiments, such movement of the carriage **120** can actuate the corresponding micro-switch **112** and signal to the

dispensing system **10** that the container **200** is seated within the carriage assembly **100**.

(74) FIG. **21** depicts cross-sectional views of the container **200** seated within the carriage **120**. Several distances and dimensions are highlighted relating to the carriage inlet **134**, the carriage outlet **136**, the inlet **224**, and the outlet **226**. These distances and dimensions include a distance **D** between the central longitudinal axis of the inlet **224** and the central longitudinal axis of the outlet **226** and an outlet valve diameter **Vd**. Also illustrated are a diameter **Y1** of the carriage inlet **134**, a diameter **Y2** of the carriage outlet **136**, and an allowable misalignment **291** between the carriage outlet **136** and the outlet **226**. The allowable misalignment **291** can define the effective difference in distances between the respective components of the carriage **120** and the container **200**, while still enabling a beverage dispensing process to take place. As seen in more detail in FIG. **22**, the carriage outlet **136** can include an outlet receiver rim **136A** (also known as a seal) that is sized to fit within the outlet **226**. The outlet receiver rim **136A** and the outlet **226** can together form a sealing surface such that an additive can be dispensed from the outlet **226** during a beverage dispensing process without concern for leaks or inaccurate dosages. If an ingredient container were seated on the carriage **120** and the dimensions of that ingredient container were such that the outlet receiver rim **136A** was not properly received in the container outlet **226**, then a beverage dispensing process could potentially be compromised. In some aspects, the rim **136A** (or seal) can be between about 7 mm to 8 mm in diameter.

(75) In certain embodiments, the distance **D** between the inlet and outlet can be between about 11 mm and 15 mm, and in some embodiments it can be about 13 mm. **Vd** can be between about 8 and 11 mm, and in some it embodiments can be about 9.5 mm. **Y1** can be between about 7.7 mm and 8.1 mm, and in some embodiments it can be about 7.91 mm. **Y2** can be between about 7.5 mm and 7.9 mm, and in some embodiments it can be about 7.70 mm. The allowable misalignment **291** can be between about 0.3 mm and 0.6 mm, and in some embodiments it can be about 0.5 mm.

(76) When the ingredient container **200** is properly seated in the carriage assembly **100**, a beverage dispensing process can occur using the stored additive. A user can select their beverage preferences, specifying details including volume, carbonation level, additive type, additive amount, and more. When the selections are received by the dispensing system **10**, a beverage can be dispensed with the selected characteristics.

(77) If an additive is desired, air or another gas, including carbon dioxide, nitrogen, oxygen, and the like, can be pumped through the gas line **116** and into the interior of the container body **250** through the inlet port **142** in the carriage **120** and through the inlet valve **244D** in the container **200**. The resulting increase in pressure within the ingredient container **200** can cause the outlet valve **226D** to open and additive to dispense through the outlet **226** and the outlet port **244**, into a drink container, such as the drinking glass **26** depicted in FIG. **1**. In embodiments where the additive is a fluid, the additive can be dispensed at a certain dispensing flowrate **F** under a certain pressure. For example, in some embodiments, the dispensing flow rate **F** can be between about 1 mL/sec and 4 mL/sec. In other embodiments, the dispensing flow rate **F** can be about 2 mL/sec. A base liquid, such as carbonated water, can also be dispensed from the fluid outlet **114** such that the base liquid and the additive combine in the drinking glass **26**.

(78) In an exemplary embodiment, the carriage assembly **100** and two ingredient containers **200** can be arranged to minimize a distance between the fluid outlet **114** of the carriage assembly **100** and the outlets **226** of the ingredient containers **200**. A bottom perspective of this arrangement is illustrated in FIG. **22**. Although the outlets **226** and the fluid outlet **114** can be distinct, the distance between each outlet **226** (and the outlet port **244**, in turn) and the fluid outlet **114** can be minimized as a result of the overall carriage assembly **100** configuration. The minimization of distance can arise as a result of the position of each outlet **226** on the respective ingredient containers **200**, located on a minor axis **B-B** (not shown). When the ingredient containers **200** are received in the carriage assembly, the containers **200** can be positioned such that each outlet **226** is centrally located and close to the fluid outlet **114**, which can extend between the two carriages **120**, as shown

above, for example, in FIG. 18.

(79) This minimized distance can allow for a variety of drink containers to be placed beneath the carriage assembly **100** and to receive a beverage while also minimizing splashing and overall mess. For example, several circles indicative of a scale are shown in FIG. 19, and they can represent, in order of smallest to largest, a narrow water bottle circumference α , a highball glass circumference β , a Collins glass circumference γ , a mason jar circumference δ , and a pint glass circumference ϵ . These circumferences are meant to illustrate the variety of drinkware usable with the beverage dispensing system **10** as a result of the arrangement of the containers **200** within the carriage assembly.

(80) During a dispensing procedure, accurate dosages can be important to the creation of a beverage and can affect the quality of the resulting product. This accuracy can be affected by a number of parameters, each introduced and described above, including opening pressure P_o , closing pressure P_c , the outlet valve diameter V_d of an outlet valve, and the slit length L_s on the outlet valve.

(81) Each of these parameters can affect an overall accuracy of the ingredient container **200** during a beverage dispensing process. For example, if the opening pressure P_o and/or the closing pressure P_c are too low, minor fluctuations in the internal pressure of the ingredient container **200** during a dispensing process, such as those associated with normal tolerance levels of the beverage dispensing device **10**, could contribute to inaccurate dispensing of an additive. Conversely, if the opening pressure P_o and/or the closing pressure P_c are too high, the additive could be dispensed in an extreme manner, resulting in excess spray of the additive and also resulting in inaccurate dispensing.

(82) As explained above, the opening pressure P_o is the pressure required to open the outlet valve **226D** and permit fluid to flow therethrough. Once the outlet valve **226D** is open and fluid is dispensed, the built-up pressure will taper off and decrease over time. Eventually, the pressure will reach a value that is too low to keep the outlet valve **226D** open. This lower limit is the closing pressure P_c . The difference ΔP between the opening pressure P_o and the closing pressure P_c can be optimized so as to not be either too great or too small, as this can affect the overall dosing accuracy during flavoring. The overall structure of the outlet valve, including its size, shape, and material, can alter the value of the opening pressure P_o and closing pressure P_c , which can affect performance of the ingredient container **200**. For example, if the difference ΔP is too small, minor fluctuations during a dispensing procedure could cause the outlet valve **226D** to prematurely open or close. If the difference ΔP is too large then the outlet valve **226D** could have trouble closing once opened, which could result in an inability to add small doses of an additive.

(83) If the opening pressure P_o is too high, the dispensing of fluid can become explosive, unmeasured, and/or unpredictable during dispensing, which can result in an overall loss of dosing accuracy. If opening pressure P_o is too low, minor fluctuations or disturbances could lead to leaking and accidental discharge of an additive, which could also result in an overall loss of dosing accuracy. If the closing pressure P_c is too high, especially relative to the opening pressure P_o (which would result in a small difference ΔP), then the window at which the outlet valve **226D** is open would shrink drastically, which can result in a temperamental valve that is only able to open at a small pressure window. Conversely, if the closing pressure P_c is too low, then the outlet valve **226D** will be open for too long as an additive is dispensed, which can also lead to an overall loss of dosing accuracy as the outlet valve **226D** could fail to close in a precise manner, leading to over-dosing of the additive. Accordingly, the opening pressure P_o and closing pressure P_c can be optimized to result in accurate dosing.

(84) Additionally, the outlet valve diameter V_d and the slit length L_s —values affecting the dimensions of the outlet **226** and the outlet valve **226D**—can effect dosing accuracy if they are too large or small. Forcing an additive out of a too-small or too-large slit **226H** or outlet **226** can affect process timing and overall dosing, thereby affecting the accuracy of the dispensing process.

(85) These values can vary depending upon the manufacturing process, materials, quality, etc. of the ingredient container **200**. Together, these values can contribute to a so-called Dosing Accuracy (DA) value, which can be used to rate the quality of an ingredient container **200**. In general, a given outlet valve on the ingredient container **200** can have a maximum potential in terms of a DA value, such as being able to accurately doze an additive, having a low minimum dose threshold for precision dosing, etc. This maximum potential can be limited, in some embodiments, by a quality of an inlet valve on the ingredient container **200**. For example, for a given outlet valve, a quality inlet valve will mean that the maximum potential of the ingredient container **200** can be achieved or at least nearly achieved. However, for the same outlet valve, a poor inlet valve can result in a large drop-off in performance from the outlet valve's potential.

(86) The DA value can be expressed by the following formula:

$$(87) DA = \left[\frac{(Po - Pc)}{(Vd - Lz)} \right]$$

(88) The individual values of these variables can vary, however their relationship according to the DA formula can provide a simple way to compare the quality of valves. Table 1 lays out several example values according to various designs, with each having the same outlet valve diameter D. It has been discovered that, according to the above formula, a valve having a DA value of 100 or less can accurately dose an additive in order to create a consistent beverage product. Each of the provided examples yields a DA value of 100 or less, with the exception of Examples 16 through 21. These examples pertain to valves which do not provide the ability to accurately dose an additive, likely due to the combination of properties of the valve, including the higher slit length Ls and higher difference ΔP between the opening and closing pressures Po, Pc.

(89) TABLE-US-00001 TABLE 1 Examples

Example	Po	Pc	Vd	Ls	ΔP	DA
1	483	343	9.5	2.5	140	
20	2	483	323	9.5	1.8	160
21	3	1819	1628	9.5	3.7	191
33	4	660	455	9.5	1.8	205
27	5	665	450	9.5	3.7	215
37	6	483	267	9.5	3.7	216
7	660	409	9.5	2.5	251	
36	8	483	224	9.5	4.7	259
55	9	483	218	9.5	5.1	265
60	10	1448	1163	9.5	1.8	284
37	11	960	640	9.5	2.5	320
46	12	660	318	9.5	3.7	342
59	13	1427	1019	9.5	2.5	409
59	14	848	419	9.5	3.7	429
74	15	660	208	9.5	4.7	452
95	16	660	191	9.5	5.1	469
107	17	1405	724	9.5	3.7	681
117	18	2212	1483	9.5	3.7	729
125	19	4575	3759	9.5	3.7	816
140	20	1379	305	9.5	4.7	1074
226	21	4234	2593	9.5	3.7	1641
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(90) The DA factor, in some embodiments can be less than 100 according to the above formula, and it could fall more specifically between about 40 and 70. In further embodiments, the DA factor can be about 55. In systems with a DA factor that is less than 100, beverage making processes can accurately dose an additive to within fractions of a mL. For example, an amount of additive, such as a fluid, dispensed during a process can be between about 1.6 mL and 2.0 mL, and in some embodiments can be about 1.8 mL. This volume of fluid can be dispensed after gas is pumped into the container for a predetermined time period, such as about 140 ms. Importantly, the amount of fluid dispensed by a container can be proportional to a difference between the opening and closing pressures of a given valve.

(91) As indicated above, the various properties of the valve can vary. In certain exemplary embodiments, the valve has an opening pressure Po that is about 300 mmH.sub.2O or greater, and more preferably is about 400 mmH.sub.2O or greater, or even 600 mmH.sub.2O or greater; a closing pressure Pc that is less than the opening pressure Po but that is about 100 mmH.sub.2O or greater, and more preferably is about 300 mmH.sub.2O or greater, or even, in some embodiments, 400 mmH.sub.2O or greater; a pressure differential (ΔP) that is in range of about 200 mmH.sub.2O to 500 mmH.sub.2O, and more preferably is about 300 mmH.sub.2O to 400 mmH.sub.2O, and even more preferably is about 340 mmH.sub.2O; and an outlet valve diameter Vd in a range of about 5 mm to 15 mm. In certain embodiments, Vd can be about 7 mm to 13 mm, and more preferably about 9.5 mm; a slit length Ls in a range of about 1 mm to 5 mm, and more preferably is about 3.7 mm.

(92) Certain illustrative implementations have been described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the systems, devices, and

methods disclosed herein. One or more examples of these implementations have been illustrated in the accompanying drawings. Those skilled in the art will understand that the systems, devices, and methods specifically described herein and illustrated in the accompanying drawings are non-limiting illustrative implementations and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one illustrative implementation may be combined with the features of other implementations. Such modifications and variations are intended to be included within the scope of the present invention. Further, in the present disclosure, like-named components of the implementations generally have similar features, and thus within a particular implementation each feature of each like-named component is not necessarily fully elaborated upon.

(93) Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

(94) One skilled in the art will appreciate further features and advantages of the invention based on the above-described implementations. Accordingly, the present application is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated by reference in their entirety.

Claims

1. An ingredient container, comprising: a container body having a hollow interior and an opening leading into the hollow interior; a cap positioned over the opening in the container body and including an inlet port and an outlet port, wherein a cross-section of the cap extending substantially perpendicular to a central axis of each of the inlet port and the outlet port has a shape that is a substantially circular triangle.
2. The ingredient container of claim 1, wherein the cap has an outer sidewall defining the shape of the cross-section and having first, second, and third walls.
3. The ingredient container of claim 2, wherein the first wall has a length that is greater than a length of each of the second and third walls.
4. The ingredient container of claim 2, the first wall is substantially planar, and the second and third walls are convex.
5. The ingredient container of claim 1, wherein the cap has a base wall with the inlet and outlet ports therein, and an outer sidewall surrounding the base wall, the outer sidewall having first and second shoulders projecting upward from the base wall.
6. The ingredient container of claim 2, wherein the cap includes a base wall having the inlet and outlet ports formed therein, and wherein the base wall includes a circular cavity formed therein at a mid-portion thereof.
7. The ingredient container of claim 6, wherein the inlet and outlet ports are positioned within the circular cavity.
8. An ingredient container, comprising: a container body having a hollow interior and an opening leading into the hollow interior; a lid coupled to the container body and covering the opening, the lid including an inlet leading from an exterior environment to the hollow interior and an outlet leading from the hollow interior to the exterior environment, the lid having a substantially triangular shape, wherein the lid is configured to couple to a beverage dispenser having a complementary substantially triangular receiver.

9. The ingredient container of claim 8, wherein the lid includes a lid cover movable between an open position and a closed position, the lid cover being configured to close the inlet and the outlet when in the closed position.
10. The ingredient container of claim 8, wherein the lid includes a substantially circular inlet collar disposed around the inlet and a substantially circular outlet collar disposed around the outlet, the inlet collar and the outlet collar each protruding away from the lid.
11. The ingredient container of claim 10, wherein a height of the inlet collar is greater than a height of the outlet collar.
12. The ingredient container of claim 10, wherein the lid includes a lid cover movable between an open position and a closed position, the lid cover including a substantially circular inlet cover collar and a substantially circular outlet cover collar, the inlet collar being configured to encircle the inlet cover collar and the outlet collar being configured to encircle the outlet cover collar when the lid cover is in the closed position.
13. The ingredient container of claim 12, wherein the lid cover includes a central plug disposed within the outlet cover collar and configured to be received by the outlet.
14. The ingredient container of claim 8, wherein the lid has an outer sidewall defining the substantially triangular shape.
15. The ingredient container of claim 14, wherein the sidewall comprises first, second, and third walls, the first wall having a length that is greater than a length of each of the second and third walls.
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